Enhancing Geostationary Supercooled Water Cloud Identification Using Hyperspectral Infrared Sounder Measurements

Yi Chen, *Student Member, IEEE*, Tingting Zhou, *Member, IEEE*, Zhixin Yang, Bin Guo, Miao Cai, Mohammed A. A. Al-Qaness, Fei Zhang, *Member, IEEE*, Artur Pedro Madal, and Haoyang Fu

***Abstract*—Supercooled water clouds (SWCs) play an important role in cloud radiative effects, precipitation processes, and aviation safety. Accurate SWC discrimination from passive geostationary observations remains difficult because SWCs are liquid clouds under subfreezing conditions and are influenced by atmospheric background conditions. Although reanalysis products such as ERA5 can provide useful atmospheric background fields, their latency limits near-real-time monitoring. To address this issue, this study investigates whether same-platform hyperspectral infrared observations from FY-4B/geostationary interferometric infrared sounder (GIIRS) can provide effective atmospheric-state constraints for SWC identification without relying on reanalysis inputs. A deep-learning framework, termed graph-based spectral ensemble network (GSE-Net), is developed by combining FY-4B/advanced geostationary radiation imager (AGRI) and GIIRS hyperspectral infrared observations for SWC identification. Within GSE-Net, spectral graph compressor (SGC) refines and compresses GIIRS spectra, while Bi-FiLM bidirectionally modulates spectral and AGRI scene features before fusion. Using CALIPSO as the reference, the proposed framework achieves a mean SWC *F*1 score of 89.33%, improving upon AGRI-only retrievals by 10.41 percentage points and outperforming AGRI+ERA5. In addition, the SGC further improves SWC identification performance over directly using uncompressed GIIRS observations. The results demonstrate that geostationary hyperspectral infrared observations can provide useful atmospheric-state constraints for SWC identification and serve as a practical alternative for near-real-time applications.**

***Index Terms*—Cloud phase (CLP), FY-4B/advanced geostationary radiation imager (AGRI), FY-4B/geostationary interferometric infrared sounder (GIIRS), spectral compression, supercooled water clouds (SWCs), thermal infrared.**

I. Introduction

CLOUD thermodynamic phase is a key variable in cloud remote sensing and cloud physics studies because it affects the radiative differences between liquid and ice clouds and also underpins the retrieval of related cloud properties [1], [2]. Among cloud thermodynamic phases, supercooled water clouds (SWCs) are clouds whose droplets remain liquid at temperatures below 0 °C [3], are widespread in the atmosphere [4], and are important for cloud radiative effects [5], [6], precipitation processes [7], and aviation weather hazard monitoring [8], [9]. Passive imagers on geostationary satellites provide high-temporal frequency and broad spatial coverage, which offers a practical basis for sustained SWC monitoring [10], [11]. However, SWC identification from passive observations remains challenging because the relevant radiative signals are jointly influenced by cloud-top temperature, optical depth, effective particle size, and atmospheric state, leading to substantial uncertainty in complex cloud scenes [12], [13].

Existing methods for cloud thermodynamic phase identification from geostationary satellites rely mainly on multichannel imager radiances and their derived features, including traditional approaches based on multichannel threshold tests and empirical feature combinations [14], [15], [16], as well as machine-learning- and deep-learning-based approaches [17], [18], [19], [20], [21], [22], [23], [24], [25]. For the more specific task of SWC identification, previous studies have either conducted direct detection using passive satellite or radiometer observations [14], [15], [26], [27] or characterized SWCs through cloud-top phase classification [23], [24] and retrieval of the supercooled liquid fraction in mixed-phase clouds [25]. However, for SWC identification, the key issue is not merely the general discrimination between liquid and ice phases, but determining whether liquid-cloud droplets can remain liquid under subfreezing conditions [3]. Under passive observations, this problem appears both as the discrimination between SWCs and ice clouds (ICs) and as the further subdivision between warm water clouds (WWCs) and SWCs within liquid clouds [3], [12], [13]. Existing studies have shown

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Yi Chen, Tingting Zhou, Zhixin Yang, Bin Guo, and Haoyang Fu are with the College of Physics and Electronic Information Engineering, Zhejiang Normal University, Jinhua 321004, China, and also with the China-Mozambique "Belt and Road" Joint Laboratory on Smart Agriculture, Zhejiang Normal University, Jinhua 321004, China (e-mail: fuhaoyang@zjnu.edu.cn).

Miao Cai is with CMA Cloud-Precipitation Physics and Weather Modification Key Laboratory (CPML), CMA Weather Modification Centre, China Meteorological Administration, Beijing 100081, China.

Mohammed A. A. Al-Qaness is with the College of Physics and Electronic Information Engineering, Zhejiang Normal University, Jinhua 321004, China.

Fei Zhang is with the College of Geography and Environmental Sciences and China-Mozambique "Belt and Road" Joint Laboratory on Smart Agriculture, Zhejiang Normal University, Jinhua 321004, China.

Artur Pedro Madal is with China-Mozambique "Belt and Road" Joint Laboratory on Smart Agriculture, Zhejiang Normal University, Jinhua 321004, China, and also with the Faculty of Agronomy and Forest Engineering, Eduardo Mondlane University, Maputo 1100, Mozambique.

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TABLE I
SUMMARY OF DATASETS AND FEATURE INFORMATION USED IN THIS STUDY

Category	Data source	Features	Spatial resolution	Time resolution
Input	**FY-4B/AGRI**	Thermal infrared BTs and BTDs;	4 km	15 min
	FY-4B/GIIRS	Long-wave hyperspectral BTs;	12 km	2 h
Label	**CALIPSO L2 C-Pro**	Cloud-phase labels; quality flags for quality control;	5 km	Overpass
Auxiliary	**FY-4B geometry**	Satellite zenith angle (SAZ);	4 km	Static
	C3S land-cover	Annual land-cover class;	300 m	Yearly

that multichannel imager radiances and their derived features can provide information relevant to supercooled liquid water [14], [15], [23], [24], [25], [26], but these signals are not governed by cloud phase (CLP) alone and are also affected by cloud-top temperature, optical thickness, particle size, and atmospheric state [12], [13]. Because SWC identification is fundamentally linked to whether liquid clouds occur under subfreezing conditions, multichannel imager radiances and their derived features alone usually cannot provide sufficiently strong and stable atmospheric background constraints [3], [28].

In passive satellite cloud retrievals, temperature and humidity profiles from reanalysis are often used as auxiliary background fields together with imager observations to compensate for the limited characterization of atmospheric state provided by imager radiances alone [23], [29], [30]. The fifth-generation atmospheric reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ERA5), for example, can provide continuous and relatively high-quality atmospheric temperature and humidity background information [31], but its updates still typically lag real time by about five days [32]. For near-real-time SWC identification, this multiday delay motivates the use of satellite observations with hour-scale refresh characteristics. Under this requirement, the geostationary interferometric infrared sounder (GIIRS), carried on the same platform as the advanced geostationary radiation imager (AGRI), provides a new possibility for introducing same-platform hyperspectral infrared observations. GIIRS is highly sensitive to atmospheric temperature and humidity structure [33], [34] and has been used for atmospheric temperature and humidity profile retrieval and product evaluation [35], [36], [37], cloud detection [38], and AGRI–GIIRS synergistic retrieval of cloud macrophysical properties under all-day conditions [29]. Compared with external reanalysis background fields, the same-platform synergy between AGRI and GIIRS offers advantages in terms of observation time, viewing geometry, and scene consistency, and this configuration is also beneficial for consistent infrared constraints and quantitative applications [39]. Previous work has shown that, in the retrieval of cloud macrophysical properties driven primarily by AGRI infrared observations under all-day conditions, introducing GIIRS longwave observations can achieve performance close to that obtained by using ERA5 temperature and humidity profiles [29]. Therefore, GIIRS has the potential to provide same-platform auxiliary observations for SWC identification based on hour-scale satellite measurements.

However, hyperspectral infrared observations usually involve a large number of channels, high interchannel correlation, and substantial redundancy when used as uncompressed spectral inputs. Directly using high-dimensional GIIRS longwave vectors increases the input dimensionality and computational burden, and may also make it harder to stably exploit the useful spectral information in GIIRS longwave observations for near-real-time SWC identification. Existing studies usually improve the usability of GIIRS observations through channel screening [40], [41], [42], [43], dynamic or adaptive channel selection [44], [45], and data assimilation or observational-thinning strategies [46], [47], [48]. For this study, the more critical issue is how to transform the high-dimensional and redundant GIIRS longwave vectors into a compact representation suitable for near-real-time SWC identification.

To reduce the dependence of near-real-time SWC identification on external reanalysis background fields such as ERA5, this study proposes a graph-based spectral ensemble network (GSE-Net), a supervised learning framework based on same-platform FY-4B/AGRI and GIIRS observations. In this study, near-real-time refers to the use of FY-4B satellite observations with hour-scale refresh characteristics instead of delayed reanalysis fields during inference. The framework uses AGRI multichannel thermal-infrared observations as the primary cloud-top radiometric input and introduces GIIRS hyperspectral infrared observations as the same-platform auxiliary observational source. To address the high dimensionality, strong spectral correlation, and redundancy of uncompressed GIIRS vectors, a spectral graph compressor (SGC) is designed to perform local residual refinement using actual-wavenumber neighborhoods and then transform selected GIIRS longwave dimensions into a 32-D spectral embedding for efficient fusion with AGRI features. Before fusion, Bi-FiLM allows the GIIRS spectral and AGRI scene features to condition each other rather than relying on direct concatenation. Using CALIPSO C-Pro cloud-phase labels as the reference, the experiments evaluate the contribution of the selected GIIRS longwave input and SGC spectral compression to SWC identification, and further assess the consistency of the proposed framework with active-sensor reference labels and operational cloud-phase products.

II. DATA

A. Data Sources

This study focuses on the FY-4B/GIIRS coverage over China and adjacent areas. The dataset covers the period from July 2022 to June 2023, spanning all four seasons and both daytime and nighttime conditions. To construct a reliable all-day near-real-time SWC identification model, three primary data sources are used synergistically:

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1) FY-4B AGRI thermal-infrared brightness temperatures (BTs) and BT differences (BTDs) serve as the primary predictors characterizing cloud-top radiometric properties; 2) FY-4B GIIRS longwave BTs serve as supplementary observations providing atmospheric-state information related to temperature and humidity structure; and 3) the CALIPSO L2 Cloud Profile product serves as the reference source of cloud-phase labels. Auxiliary information consists of satellite zenith angle (SAZ) and annual land-cover data from the Copernicus Climate Change Service (C3S) global 300-m land-cover product, which are used to represent viewing geometry and underlying surface differences, respectively [49], [50]. Detailed specifications of these datasets are summarized in Table I.

1) FY-4B AGRI: AGRI onboard FY-4B provides high-temporal-resolution multispectral observations. Seven thermal-infrared channels spanning 6.25–13.3 μm are used in this study. Their Level-1 measurements are converted to BTs using the calibration relationship provided in the AGRI product, and the resulting BTs together with derived BTDs are used as radiometric predictors for cloud-phase classification. Full-disk observations are available every 15 min at a nadir spatial resolution of 4 km. These thermal-infrared predictors contain information relevant to CLP and maintain a consistent feature representation across daytime and nighttime conditions, which is important for all-day cloud-phase identification.

2) FY-4B GIIRS: GIIRS onboard FY-4B is a geostationary hyperspectral infrared sounder that provides continuous observations over China and adjacent regions. FY-4B GIIRS includes a longwave band covering 680–1130 cm^{-1} and a midwave band covering 1650–2250 cm^{-1}. In this study, only the longwave band is used because it provides a stable thermal-infrared signal for both daytime and nighttime conditions, which is consistent with the all-day design of this cloud-phase identification framework. These longwave observations have also been widely used for atmospheric temperature and humidity retrieval and can provide useful spectral information for SWC discrimination [33], [37]. The native FY-4B/GIIRS Level-1 measurements are spectral radiances. In this study, these radiances are first converted to channel BTs using the radiance-to-BT calibration relationship provided with the product, and all GIIRS inputs used in the following experiments refer to the converted longwave BTs unless otherwise stated. FY-4B GIIRS completes one full observation cycle every 2 h over its observation region covering China and adjacent areas. The nadir spatial resolution of FY-4B GIIRS is about 12 km.

3) CALIPSO Lidar Cloud Profiles: The CALIPSO Level 2 Cloud Profile product (C-Pro, Version 4.51) is used as the reference label source in this study. This product is generated from observations from the Cloud-Aerosol Lidar with Orthogonal Polarization (CALIOP) together with ancillary meteorological information and provides 5-km cloud profile information along the satellite track. In this study, reference labels are constructed using the phase information encoded in the atmospheric volume descriptions together with the temperature profiles provided in C-Pro. As an active lidar product, C-Pro provides cloud profile data with vertical resolution and is sensitive to optically thin cloud layers. After spatio-temporal collocation and quality screening, the matched C-Pro data are used to assign a reference label to each footprint.

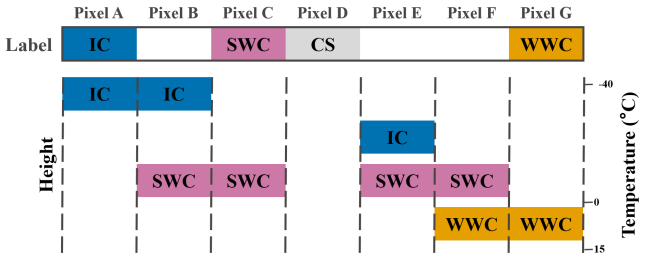


Fig. 1. Example of converting CALIPSO vertical CLP profiles into one categorical label for each footprint. Dashed lines separate adjacent footprints. Colored blocks denote the native vertical phase classifications, and the top strip shows the resulting footprint labels.

4) Ancillary Data: SAZ is included to characterize viewing-geometry differences across the full disk. In addition, differences in surface emissivity among land-cover types can affect thermal-infrared observations and increase the difficulty of cloud detection over land. To account for this surface dependence, land-cover data from the C3S global 300-m product are used as an auxiliary categorical feature. This product provides annual land-cover classes defined according to the land-cover classification system (LCCS) [49], [50]. Because land-cover types are generally stable over short time scales, the annual land-cover map for the corresponding year is used and matched to each footprint using nearest-neighbor geographic coordinates.

B. Dataset Construction

To construct a high-quality dataset, FY-4B AGRI/GIIRS observations are strictly collocated with the CALIPSO C-Pro product in space and time, followed by quality control. The main steps are described below.

1) Label Definition: Reference labels are derived from the CALIPSO C-Pro product by jointly using the atmospheric volume descriptions, which contain vertically resolved feature type and cloud ice–water phase information, and the temperature profiles reported in the product. To ensure label reliability, only footprints with medium or high ice/water phase quality assessment (QA) are retained for phase-labeled cloud cases. In particular, a footprint is classified as follows.

1) SWC if only one liquid-cloud layer is identified and all corresponding liquid-cloud bins are below 0 °C.
2) WWC if only one liquid-cloud layer is identified and all corresponding liquid-cloud bins are at or above 0 °C.
3) Ice cloud (IC) if only one ice cloud layer is identified.
4) Clear sky (CS) if no cloud feature is identified in the collocated C-Pro profile. Footprints containing multiple cloud layers, unknown phase classifications, or insufficient phase QA are discarded. Each collocated CALIPSO vertical profile is converted into a single categorical label using the four-class scheme of CS, IC, WWC, and SWC.

To illustrate this profile-to-label conversion, Fig. 1 presents an example of CALIPSO vertical CLP profiles and the resulting footprint labels along the satellite track.

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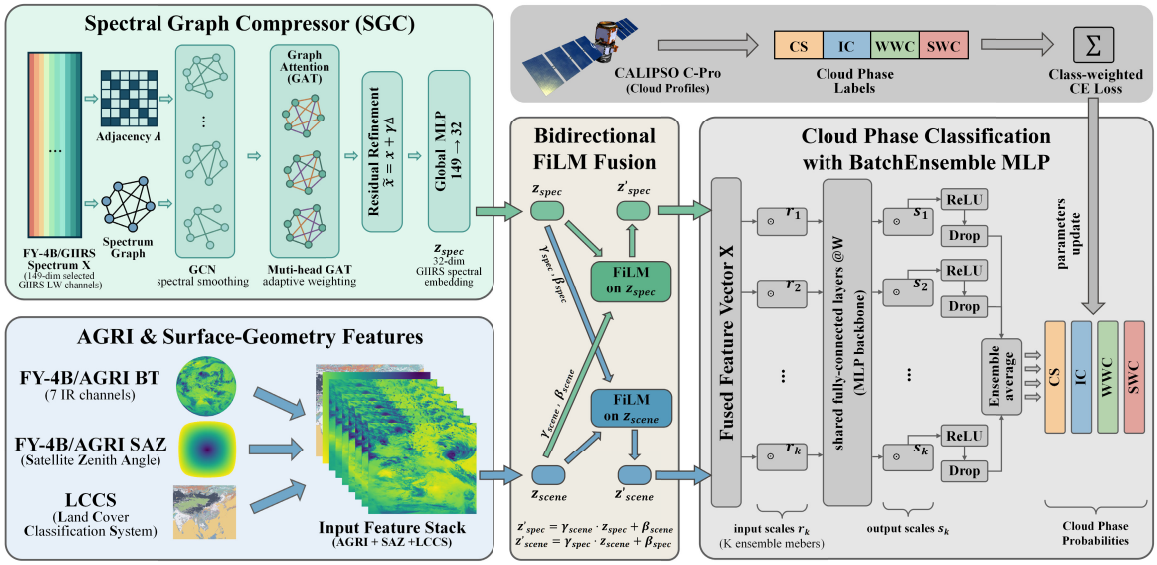


Fig. 2. Overall architecture of the proposed GSE-Net. FY-4B AGRI BTs, BTDs, SAZ, and LCCS provide scene descriptors and viewing-geometry information. The SGC extracts compact spectral features from a selected 149-D GIIRS longwave vector [51] through residual refinement. The spectral and scene embeddings are fused through a bidirectional FiLM module and classified by a shared parameter-efficient ensemble MLP, optimized with a class-weighted cross-entropy loss using CALIPSO C-Pro CLP labels.

2) Spatio-Temporal Matching: Accurate spatio-temporal matching between FY-4B observations and CALIPSO C-Pro profiles is essential for constructing reliable collocated footprints. AGRI and GIIRS observations are collocated with each CALIPSO footprint within a unified matching procedure. Sensor-specific temporal and spatial criteria are applied to account for their different observing cadences and sampling geometries. AGRI provides the primary cloud-scene observations. For each CALIPSO footprint, the nearest valid AGRI pixel is retained when the absolute time difference does not exceed 15 min, and the geographic distance is below 0.1°. GIIRS is a sounding instrument that provides auxiliary hyperspectral information related to the atmospheric temperature and humidity structure at discrete FOV centers within the scanned region. For each CALIPSO footprint, we retain the temporally nearest GIIRS observation within 2 h whose nearest valid FOV lies within 0.5°, using spatial distance only to resolve equal time differences. Sensitivity experiments confirm the robustness of model performance to the matching criteria, with detailed results provided in Supplementary Section S2.

3) Quality Control and Data Splitting: To reduce label ambiguity caused by mixed or transitional footprints near cloud boundaries, only sequences of at least five consecutive CALIPSO footprints with identical phase labels are retained. From each valid sequence, the first and last two footprints are discarded so that only the central high-confidence footprints are kept. Because the dataset spans only one year (July 2022 to June 2023), splitting the samples by season would substantially reduce the diversity of the training set. A purely random split at the footprint scale would place neighboring footprints from the same CALIPSO overpass into different subsets, leading to information leakage and overly optimistic performance estimates. To avoid this problem, all footprints from the same CALIPSO overpass within the FY-4B/GIIRS coverage over China are kept together and assigned to the same fold, and the matched overpasses are randomly divided into five mutually exclusive folds. The resulting dataset comprises 998 937 collocated footprints within the FY-4B/GIIRS coverage over China, including 514 347 CS samples (51.489%), 418 855 IC samples (41.930%), 20 528 WWC samples (2.055%), and 45 207 SWC samples (4.526%).

III. Methodology

The proposed GSE-Net integrates an SGC for high-dimensional spectral compression and a parameter-efficient ensemble classifier for cloud-phase discrimination. This architecture uses a selected 149-dim FY-4B/GIIRS longwave vector following [51], in which radiative transfer simulations and validation based on Infrared Atmospheric Sounding Interferometer (IASI) observations were used to retain channels that are sensitive to cloud microphysical properties and have relatively stable data quality. Although this selected vector is already much smaller than the full longwave GIIRS vector, it remains high-dimensional and spectrally redundant for direct model input and near-real-time application, and its compressed spectral representation is therefore fused with imager-derived contextual information. The network takes FY-4B AGRI thermal-infrared BTs and BTDs, SAZ, LCCS, and FY-4B GIIRS longwave BTs as inputs, while collocated CALIPSO C-Pro data are used to construct the reference labels. The overall architecture of GSE-Net is shown in Fig. 2.

A. Spectral Input Configurations and Graph Compressor

In the default setting, the GIIRS spectral input is a selected 149-D longwave vector for cloud-phase discrimination. These channels are selected following the channel screening procedure described in [51], which combines radiative transfer simulations to identify channels sensitive to cloud microphysical properties with validation based on IASI observations to exclude channels with large calibration uncertainties.

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Because adjacent spectral dimensions exhibit structured correlations in wavenumber space, a SGC is introduced to transform the selected 149-D vector into a 32-D spectral embedding. Rather than treating the selected channels as an unstructured vector, we represent them as a spectral graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node corresponds to one channel and edges connect nearby channels in wavenumber space. The graph is constructed using the actual center wavenumbers of the selected channels, with connected channels weighted by $A_{ij} = 1/(1 + |\nu_i - \nu_j|)$. To incorporate self-loops and normalize neighborhood aggregation, the symmetrically normalized adjacency matrix is defined as

$$\tilde{\mathbf{A}} = \tilde{\mathbf{D}}^{-\frac{1}{2}} (\mathbf{A} + \mathbf{I}) \tilde{\mathbf{D}}^{-\frac{1}{2}} \tag{1}$$

where $\mathbf{I}$ is the identity matrix and $\tilde{\mathbf{D}}$ is the degree matrix of $\mathbf{A} + \mathbf{I}$. This graph formulation enables the model to exploit structured correlations among nearby channels instead of processing all channels independently. A two-layer graph convolutional network (GCN) is then applied to aggregate local spectral information on this graph

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{A}}\, \mathbf{H}^{(l)}\, \mathbf{W}^{(l)}\right), \quad l = 0, 1 \tag{2}$$

where $\tilde{\mathbf{A}}$ denotes the normalized adjacency matrix defined in (1) and $\sigma(\cdot)$ is the GELU activation function.

The GCN mainly aggregates information within local wavenumber neighborhoods, whereas the subsequent graph attention network (GAT) layer adaptively reweights channel interactions to refine the spectral representation. For the GAT layer, the attention coefficient e_{ij} between channel i and j is computed as

$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}^{\top}\left[\mathbf{W}\mathbf{h}_i \,\|\, \mathbf{W}\mathbf{h}_j\right]\right) + \log\left(A_{ij} + \epsilon\right) \tag{3}$$

where $\|$ denotes concatenation and $\mathbf{a}$ is a learnable weight vector. The attention weights are then normalized via softmax: $\alpha_{ij} = \exp(e_{ij}) / \sum_{k \in \mathcal{N}_i} \exp(e_{ik})$. Consequently, the feature of node i is updated by aggregating the weighted neighbors

$$\mathbf{h}_i' = \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W}\mathbf{h}_j\right). \tag{4}$$

The final node features are mapped to a channelwise residual $\Delta\mathbf{x}$, which is added to the original GIIRS spectrum as

$$\tilde{\mathbf{x}} = \mathbf{x} + \gamma \Delta\mathbf{x} \tag{5}$$

where γ is a learnable scaling factor. This residual form preserves the original spectrum and lets the local graph branch learn only the required channelwise corrections. A multilayer perceptron (MLP) then jointly encodes all refined channels into a fixed-dimensional spectral embedding $\mathbf{z}_{\text{spec}} \in \mathbb{R}^d$ with $d = 32$, thereby capturing full-spectrum relationships.

B. Feature Fusion and Ensemble Classification

To integrate spectral cues from GIIRS with thermal-infrared scene information from FY-4B AGRI, together with viewing geometry and surface type, a bidirectional FiLM (featurewise linear modulation, Bi-FiLM) module [52] is introduced.

Different from direct concatenation, this module enables the spectral and scene branches to modulate each other through learnable affine transformations. Let $\mathbf{z}_{\text{scene}}$ denote the scene embedding formed from AGRI thermal-infrared BTs, BTDs, SAZ, and LCCS. In the forward path, $\mathbf{z}_{\text{scene}}$ modulates the GIIRS spectral embedding $\mathbf{z}_{\text{spec}}$ as

$$\mathbf{z}'_{\text{spec}} = \gamma\left(\mathbf{z}_{\text{scene}}\right) \odot \mathbf{z}_{\text{spec}} + \beta\left(\mathbf{z}_{\text{scene}}\right) \tag{6}$$

where $\gamma(\cdot)$ and $\beta(\cdot)$ are the learnable functions that generate scaling and shifting terms from the conditioning embedding. This conditioning allows the spectral representation to adapt to the accompanying scene information. A symmetric path is applied in the opposite direction

$$\mathbf{z}'_{\text{scene}} = \gamma\left(\mathbf{z}_{\text{spec}}\right) \odot \mathbf{z}_{\text{scene}} + \beta\left(\mathbf{z}_{\text{spec}}\right). \tag{7}$$

The two modulated embeddings are concatenated to form $\mathbf{z}_{\text{fused}}$, which is then fed to a parameter-efficient ensemble classifier to predict CLP probabilities.

To improve predictive stability without training multiple independent models, a rank-one weight modulation strategy following BatchEnsemble [53] is adopted. In particular, the model forms an implicit ensemble of M branches that share a central weight matrix $\mathbf{W}$ while maintaining diversity through branch-specific rank-one vectors $\mathbf{r}_m$ and $\mathbf{s}_m$. The effective weight of the mth branch is written as $\mathbf{W}_m = \mathbf{W} \circ (\mathbf{s}_m \mathbf{r}_m^{\top})$. The predictive probability distribution is obtained by averaging the softmax outputs of M ensemble branches

$$p(y|\mathbf{x}) = \frac{1}{M} \sum_{m=1}^{M} \text{Softmax}\left(\text{MLP}_m\left(\mathbf{z}_{fused}; \theta_m\right)\right) \tag{8}$$

where θ_m represents the implicit efficient parameters of the mth branch. During inference, the outputs of all M branches are averaged to produce the final probability estimate. The network is trained end-to-end with a class-weighted cross-entropy loss to address class imbalance. The objective function is defined as

$$\mathcal{L} = -\frac{1}{N_{batch}} \sum_{i=1}^{N_{batch}} \sum_{c \in \mathcal{C}} w_c\, y_{i,c} \log p\left(c|\mathbf{x}_i\right) \tag{9}$$

where w_c denotes the class weight for class c, computed from the inverse class frequency raised to the power $\alpha = 0.1$. Sensitivity to α is evaluated in Supplementary Section S5.4.

C. Verification Metrics

Cloud-phase retrieval performance is evaluated using standard multiclass metrics and supplemented by binary ice–water scores for consistency with the verification metrics used for the FY-4B Level-2 CLP product [54].

1) Multiclass Metrics: Let $\mathcal{C} = \{\text{CS}, \text{IC}, \text{WWC}, \text{SWC}\}$ denote the set of four CLP classes. Standard multiclass metrics are adopted, including the overall accuracy (OA) and the $F1$ score. The primary evaluation criteria are the macroaveraged $F1$ score, denoted as $F1_{\text{macro}}$ and defined as the arithmetic mean of per-class $F1$ scores, and the class-specific $F1$ score for SWC, denoted as $F1_{\text{SWC}}$. F1_{SWC} is reported separately to emphasize the retrieval performance for the SWC class, which is particularly relevant to aviation safety and weather-modification applications.

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TABLE II

PERFORMANCE OF GSE-NET UNDER DIFFERENT INPUT SETTINGS USING CALIPSO C-PRO AS REFERENCE. METRICS ARE REPORTED AS FIVEFOLD MEAN±STANDARD DEVIATION (%)

Scheme	OA	Macro F1	$F1_{CS}$	$F1_{IC}$	$F1_{WWC}$	$F1_{SWC}$
AGRI-only	87.68±0.24	81.56±1.23	89.01±0.24	87.79±0.38	70.52±4.23	78.92±0.89
AGRI + ERA5	91.40±0.21	88.70±0.64	92.30±0.19	91.11±0.31	85.05±1.78	86.33±0.74
AGRI + GIIRS (Ours)	**92.31±0.18**	**89.53±0.84**	**92.94±0.11**	**92.27±0.35**	**83.57±2.80**	**89.33±0.64**

TABLE III

COMPARISON OF DIFFERENT CLASSIFIERS USING CALIPSO C-PRO AS REFERENCE UNDER THE AGRI+GIIRS INPUT CONFIGURATION. METRICS ARE REPORTED AS FIVEFOLD MEAN±STANDARD DEVIATION (%)

Model	OA	Macro F1	$F1_{SWC}$
Random Forest	91.66±0.27	84.81±1.04	76.38±1.52
XGBoost	91.72±0.39	87.91±1.16	85.89±0.98
CatBoost	91.52±0.44	86.70±1.17	82.81±1.09
LightGBM	91.78±0.39	87.97±1.02	85.25±0.97
FT-Transformer	91.37±0.38	87.85±1.15	86.68±1.12
SAINT	91.26±0.43	87.45±1.19	86.33±0.97
GSE-Net	**92.31±0.18**	**89.53±0.84**	**89.33±0.64**

2) Binary Ice–Water Verification: For comparison with the AGRI L2 CLP product, the four classes are aggregated into two categories, namely water (WWC + SWC) and ice (IC), while CS is excluded from this binary evaluation. Following the metric definitions reported for AGRI L2 CLP verification [54], the hit rate (HR) and Kuipers skill score (KSS) are computed from the binary ice–water contingency table. These binary scores enable direct comparison with the reported verification results of the AGRI L2 CLP product.

IV. RESULTS AND DISCUSSION

A. Contribution of GIIRS Longwave Observations

This section evaluates the contribution of same-platform GIIRS observations to SWC identification relative to an AGRI-only baseline and an ERA5-assisted reference configuration. All configurations are assessed under the same fivefold cross-validation scheme by augmenting the AGRI thermal-infrared baseline with either ERA5 temperature and relative humidity profiles at 1000, 850, 500, and 200 hPa or the collocated selected FY-4B/GIIRS longwave vector.

As shown in Table II, the AGRI-only configuration already provides strong four-class discrimination. This result indicates that FY-4B AGRI thermal-infrared BTs and derived BTDs already contain substantial radiometric information for separating CS, ice cloud, and liquid-phase cloud conditions in many matched scenes. The AGRI+ERA5 configuration improves overall OA and Macro $F1$, with Macro $F1$ increasing from 81.56% to 88.70%, while GSE-Net achieves 89.53%. For SWC, the AGRI+ERA5 configuration increases $F1$ from 78.92% to 86.33%, whereas GSE-Net achieves 89.33%. The overall improvement from ERA5 is physically expected because phase states near freezing depend not only on cloud-top radiative signatures but also on the surrounding atmospheric environment. By providing vertically resolved temperature and humidity information, ERA5 strengthens the atmospheric background constraints needed to distinguish liquid and ice cloud phases in ambiguous cases. However, the practical use of ERA5 in near-real-time applications remains limited by its latency, as ERA5 updates are typically available about five days behind real time [32].

This limitation motivates the use of GIIRS longwave BTs from the same FY-4B platform as an auxiliary observational source for cloud-phase discrimination based on hour-scale satellite observations. Overall, the SGC-based AGRI+GIIRS setting increases SWC $F1$ from 78.92% to 89.33% relative to AGRI-only and exceeds the 86.33% achieved by AGRI+ERA5 while avoiding reanalysis input during inference.

B. Comparison With Baseline Classifiers

The AGRI+GIIRS configuration is further evaluated against representative baseline classifiers to assess whether the proposed architecture improves the use of the same input information. All compared models use the same matched footprints and the same fivefold cross-validation scheme defined at the CALIPSO overpass scale, with identical fold assignments used in each comparison. The following results, averaged across the five folds, correspond to the input configuration that jointly uses FY-4B AGRI observations and the selected GIIRS longwave input. The FT-Transformer and SAINT baselines use the same SGC and Bi-FiLM modules as GSE-Net. Results for variants without these modules, which use the uncompressed 149-channel GIIRS vector, are provided in Supplementary Section S3.3. Tree-based baselines use the uncompressed 149-channel GIIRS vector because split-based training cannot jointly optimize the learnable SGC.

As shown in Table III, tree-based baselines already achieve strong performance, indicating that the collocated FY-4B AGRI predictors and GIIRS longwave inputs contain substantial discriminative information for cloud-phase retrieval. For FT-Transformer and SAINT, incorporating SGC and Bi-FiLM produces only marginal performance changes, indicating limited additional benefit for these Transformer-based classifiers. The corresponding results are provided in Supplementary Section S3.3. Among the compared learning models in this table, GSE-Net with SGC achieves the best overall performance, with particularly clear gains in Macro $F1$ and $F1_{SWC}$.

The confusion matrices in Fig. 3 provide a class-specific view of the retrieval behavior of GSE-Net. For SWC, the model achieves a precision of 88.23% and a recall of 90.46%, indicating that SWCs can be identified with both high reliability and high completeness.

C. Evaluation Against the AGRI L2 CLP Product

The AGRI L2 CLP product is a physics-based cloud-phase retrieval. According to the official product instruction, the

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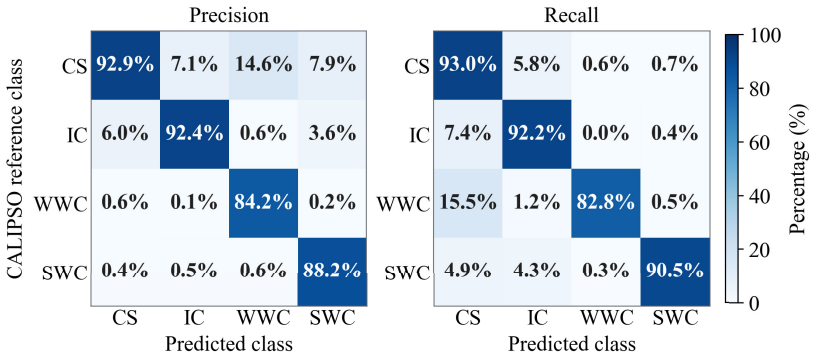


Fig. 3. Precision- and Recall-normalized confusion matrices of GSE-Net based on the pooled fivefold predictions. Rows denote CALIPSO reference classes and columns denote predicted classes. The left matrix is normalized within predicted-class columns, and the right matrix is normalized within CALIPSO-reference rows.

TABLE IV

COMPARISON BETWEEN THE AGRI L2 CLP PRODUCT AND GSE-NET USING CALIPSO C-PRO AS THE REFERENCE ON THE CLP-MATCHED DATASET (FIVEFOLD MEANS)

Method	OA	Macro F1	$F1_{SWC}$	HR	KSS
AGRI L2 CLP	0.7240	0.5645	0.4551	0.7775	0.7323
GSE-Net (AGRI-only)	0.8786	0.8183	0.7768	0.9883	0.9522
GSE-Net (AGRI+GIIRS)	**0.9228**	**0.8965**	**0.8919**	**0.9946**	**0.9730**

AGRI L2 CLP product uses observations at 7.4, 8.5, 10.8, and 12 μm, together with estimated clear-sky atmospheric radiance, clear-sky transmittance, and temperature profiles, to derive cloud microphysical information for phase discrimination. In this framework, phase-sensitive information is mainly derived from the 8.5- to 10.8-μm channels, particle-size-related information is additionally characterized using the 10.8–12-μm channels, and the 10.8-μm BT provides a constraint on cloud-top thermodynamic temperature for phase determination, especially for opaque clouds [54]. For comparison with the FY-4B AGRI L2 CLP product, the CLP retrievals are additionally matched to the footprints collocated with CALIPSO in this study. After rematching the original AGRI+GIIRS collocated dataset with the available CLP retrievals, 743 922 samples are retained for this comparison. This procedure keeps the CALIPSO C-Pro labels as the common reference while allowing the operational AGRI cloud-phase product and the proposed GSE-Net models to be evaluated on the same CLP-matched sample set for comparison using the same CALIPSO reference and metric definitions. Because the operational FY-4B AGRI L2 CLP product is available only from August 20, 2022 onward, the resulting CLP-matched subset is smaller than the full AGRI+GIIRS matchup dataset. In addition to the four-class metrics (OA and Macro $F1$), we compute the binary ice/water metrics (HR and KSS) for consistency with the AGRI L2 CLP verification definitions [54].

As shown in Table IV, the HR obtained for the AGRI L2 CLP product in our matchup evaluation is broadly consistent with the verification level reported in the official quality-assessment reports. For the two proposed GSE-Net variants, HR and KSS are already very similar under the binary formulation, making them less sensitive to the remaining performance differences. The distinction between these two configurations is therefore more clearly reflected by the four-class metrics, especially $F1_{SWC}$. In particular, $F1_{SWC}$ increases from 0.7768 for GSE-Net (AGRI-only) to 0.8919 for GSE-Net (AGRI+GIIRS), while the AGRI L2 CLP product reaches only 0.4551. Compared with GSE-Net (AGRI-only), GSE-Net (AGRI+GIIRS) provides the most meaningful improvement in the four-class evaluation, indicating the value of incorporating GIIRS longwave BT information for SWC-oriented CLP discrimination. Overall, the comparison highlights the potential of data-driven same-platform imager–sounder fusion for improving operational AGRI cloud-phase discrimination, particularly for the SWC class.

D. Component Ablation and Configuration Study

This section analyzes the effects of spectral graph compression, Bi-FiLM fusion, and BatchEnsemble under the setting that combines FY-4B AGRI predictors and GIIRS longwave inputs.

As shown in Table V, removing SGC produces the largest reductions in mean Macro $F1$ and SWC $F1$ (2.27 and 2.89 percentage points), showing that SGC more effectively extracts cloud-phase-relevant features from the high-dimensional GIIRS observations, especially for the minority cloud classes. Bi-FiLM and BatchEnsemble provide complementary improvements to the overall performance. Complete foldwise results are provided in Supplementary Section S3.4.

To further examine whether SGC improves the discrimination between IC and SWC, Fig. 4 shows the comparison of their feature distributions before and after compression. Because the original feature spaces are high-dimensional, t-distributed stochastic neighbor embedding (t-SNE) is used here only as an auxiliary tool to visualize local neighborhood structure in two dimensions [57]. To avoid overinterpreting the 2-D embeddings, the silhouette coefficient (SC), Davies–Bouldin index (DBI), and Calinski–Harabasz Index (CH) are additionally evaluated in the corresponding original feature spaces, where higher SC and CH and lower DBI indicate better class separability [58], [59], [60]. SC compares the within-class compactness of each sample with its separation from the nearest alternative class; DBI averages the worst ratio of within-class dispersion to between-centroid distance across classes; and CH is the degrees-of-freedom-normalized ratio of

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TABLE V
COMPONENT ABLATION UNDER THE AGRI+GIIRS CONFIGURATION. METRICS ARE REPORTED AS FIVEFOLD MEAN±STANDARD DEVIATION (%)

Variant	OA	Macro F1	$F1_{CS}$	$F1_{IC}$	$F1_{WWC}$	$F1_{SWC}$
GSE-Net (Ours)	**92.31±0.18**	**89.53±0.84**	**92.94±0.11**	**92.27±0.35**	**83.57±2.80**	**89.33±0.64**
w/o SGC	91.39±0.40	87.26±1.20	92.19±0.29	91.53±0.50	78.88±3.20	86.44±1.10
w/o Bi-FiLM	91.91±0.28	88.62±0.82	92.62±0.17	91.92±0.42	81.53±2.55	88.39±0.60
w/o BatchEnsemble	92.07±0.32	88.87±0.94	92.75±0.19	92.10±0.52	81.91±3.10	88.73±0.67

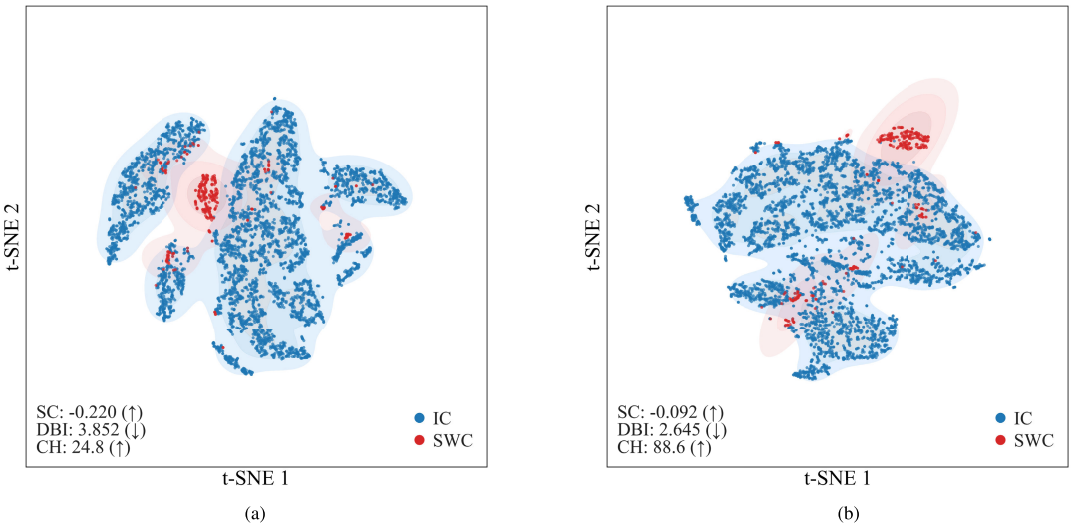


Fig. 4. t-SNE visualization of IC and SWC features before and after SGC compression. (a) Shows the feature distribution without SGC. (b) Shows the compressed feature distribution with SGC. The horizontal and vertical axes denote the two t-SNE embedding coordinates, which are used only for visualization and do not have direct physical meanings. The inset reports the SC, DBI, and CH, which are computed in the corresponding original feature spaces rather than in the 2-D embeddings; higher SC and CH and lower DBI indicate better separability.

between-class to within-class dispersion. Their mathematical definitions are provided in Supplementary Section S3.5. After SGC compression, SC improves from −0.220 to −0.092, DBI decreases from 3.852 to 2.645, and CH increases from 24.8 to 88.6, indicating improved separability between IC and SWC in the original feature space. Some overlap still remains, which is consistent with the inherent physical difficulty of distinguishing SWC from IC.

E. Robustness Assessment

To assess robustness under different observation conditions, we separately compared model performance between daytime and nighttime, across SAZ ranges, and over land and ocean. Daytime and nighttime performance is comparable, consistent with the use of longwave infrared observations that do not depend on solar illumination. At SAZ ≥ 70°, the most pronounced degradation occurs for WWC, whereas SWC $F1$ decreases more moderately. Performance over land and ocean is comparable overall, while SWC $F1$ is higher over ocean. Detailed metrics, confusion matrices, and difficult-case analyses are provided in Supplementary Section S6.

F. Case Study

An FY-4B AGRI scene acquired over eastern China at 06:30:00 UTC on January 23, 2023, is analyzed as a case study. The scene is collocated with a CALIPSO overpass and is used for along-track qualitative comparison. As shown in the true-color image in Fig. 5(a), the CALIPSO ground track intersects a frontal cloud band. The along-track profile using CALIPSO as the reference in Fig. 5(i) exhibits a pronounced transition between SWC and IC, which corresponds to the dominant confusion mode identified in Section IV-B. This case is used to qualitatively compare the AGRI+ERA5 baseline and the AGRI+GIIRS schemes in their along-track SWC diagnosis.

The spatial distribution of SWC confidence is described by the SWC margin, defined as the predicted probability of the SWC class minus the maximum predicted probability among all competing classes

$$M_{\mathrm{SWC}}(x) = P(\mathrm{SWC}|x) - \max_{c \neq \mathrm{SWC}} P(c|x) \tag{10}$$

where $P(c|x)$ denotes the predicted probability for class c. A positive margin indicates that SWC has the highest predicted probability, whereas a negative margin indicates that another class is preferred over SWC.

Fig. 5(b)–(d) shows visible differences in the SWC margin fields across the three schemes. Relative to the two baselines, the AGRI+GIIRS scheme with SGC produces stronger positive margins within the main SWC region and stronger negative margins outside it. At the same time, the transition zone between SWC and non-SWC becomes narrower, corresponding to a smaller area where the SWC margin is close to zero. This pattern suggests clearer separation between SWC and its competing classes. Between the two baselines, the ERA5-driven margin field is visually stronger than the

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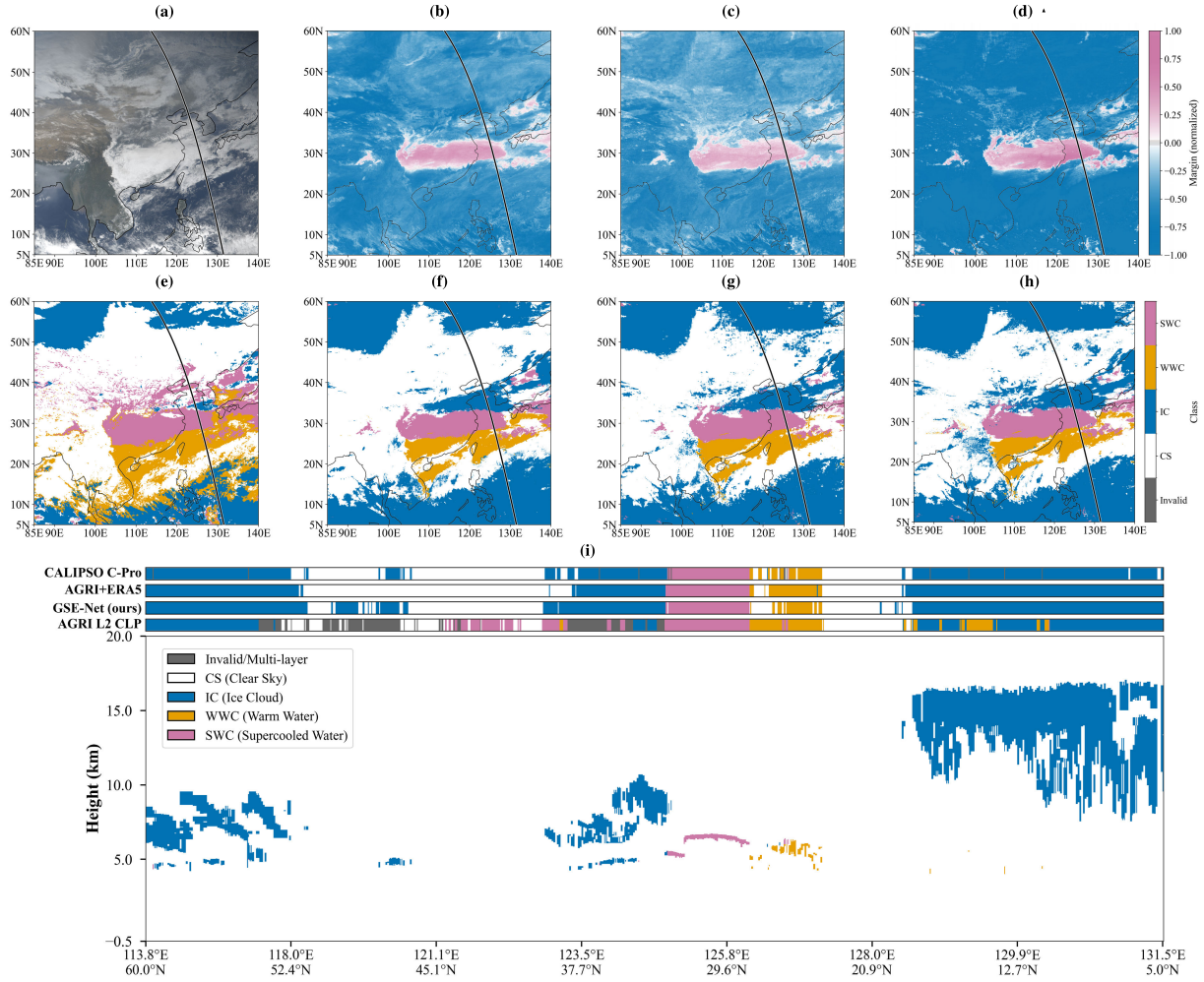


Fig. 5. Case study of cloud-phase classification for the FY-4B AGRI scene acquired over eastern China at 06:30:00 UTC on January 23, 2023. (a) Shows the true-color image together with the CALIPSO ground track. (b)–(d) SWC margin maps for the AGRI+ERA5 baseline, the AGRI+GIIRS baseline, and the AGRI+GIIRS scheme with SGC, respectively. The SWC margin is defined as the predicted probability of SWC minus the highest predicted probability among the other classes; positive values indicate preference for SWC, whereas negative values indicate preference for a non-SWC class. (e) Shows the FY-4B/AGRI L2 CLP product. (f)–(h) Predicted cloud-phase maps from AGRI+ERA5, AGRI+GIIRS, and AGRI+GIIRS with SGC. (i) Shows the CALIPSO 5-km cloud profile together with the along-track labels and predictions from the compared schemes.

uncompressed GIIRS BT baseline over much of the scene. Relative to the uncompressed GIIRS BT baseline, the AGRI+GIIRS scheme with SGC shows a more contrasted SWC margin pattern, while all three schemes still exhibit near-zero margins near transition zones and cloud boundaries.

Fig. 5(e)–(h) shows the comparison of the phase maps produced by the AGRI L2 CLP product and by the learning-based schemes, while Fig. 5(i) provides the CALIPSO C-Pro product as the along-track reference for qualitative assessment. In the SWC segment defined by the CALIPSO referenced strip in Fig. 5(i), the AGRI+ERA5 and GSE-Net strips show consistent SWC identification, and the GSE-Net result is comparable to the ERA5 baseline in the extent of the SWC segment along the track. In the IC-dominated portions of the CALIPSO-referenced strip in Fig. 5(i), the ERA5 baseline exhibits more IC to CS substitutions than GSE-Net in this case, which is consistent with a higher omission tendency for IC along the track. Relative to the AGRI L2 CLP result, the AGRI+ERA5 and GSE-Net strips show closer agreement along the track with the CALIPSO reference strip, which is consistent with the quantitative comparison in Section IV-C.

Overall, this case provides qualitative evidence that the SGC-compressed GIIRS spectral embedding can yield along-track SWC diagnosis that is qualitatively comparable to the ERA5 baseline in this case, without relying on reanalysis input. This result suggests the potential utility of geostationary observations for near-real-time aviation icing support within the FY-4B/GIIRS observation region over China and adjacent areas.

V. Conclusion

This study develops an all-day, near-real-time framework for SWC identification using CALIPSO C-Pro reference labels and collocated FY-4B AGRI/GIIRS observations. The framework combines FY-4B/AGRI multispectral thermal-infrared observations, FY-4B/GIIRS hyperspectral infrared sounding observations, and auxiliary predictors such as SAZ and land-cover class to identify cloud thermodynamic phase. The proposed GSE-Net uses an SGC to encode the selected high-dimensional GIIRS longwave vector into a compact spectral embedding, which is then fused with AGRI thermal-infrared

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features and ancillary predictors for cloud-phase classification. Evaluations using CALIPSO as the reference show that the scheme using SGC-compressed GIIRS embeddings gives the highest OA, Macro $F1$, and SWC $F1$, reaching 92.31%, 89.53%, and 89.33%, respectively, while avoiding reliance on reanalysis inputs with multiday latency such as ERA5. Under the same CLP-matched evaluation setting, the proposed learning-based framework shows higher agreement with CALIPSO References than the official AGRI L2 CLP product in both the binary ice–water verification and the four-class phase evaluation.

The current study is constrained by the limited temporal overlap between FY-4B/GIIRS and CALIPSO, which restricts the coverage of collocated training and evaluation footprints. The present label design also excludes mixed-phase clouds, and extending the label space to mixed-phase conditions remains an important direction. Future work will use next-generation active missions such as EarthCARE to expand the reference dataset and investigate model adaptation strategies, including transfer learning and retraining, to improve cross-mission consistency. MTG-S1 also carries a geostationary hyperspectral infrared sounder, but stable products are not yet available for this study. After release, we will use MTG hyperspectral infrared and imager observations to evaluate the transferability of the framework beyond FY-4B. In addition, the framework will be extended from SWC identification at the footprint scale toward the retrieval of the vertical structure of supercooled liquid water and the quantitative estimation of supercooled liquid water content and related indicators, with the aim of providing broader support for studies of cloud radiative effects, precipitation processes, and aviation meteorological applications.

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